Text As Data And Computational Measurement

Empirical Methods — Week 12

Teaching deck draft

Week 12 question

- How can raw text, images, audio, video, or multimodal records become economic variables?
- What latent object is being measured?
- Why does the raw modality contain signal about that object?
- How do measurement errors affect the descriptive, causal, or structural claim?

The lecture is about unstructured data as a measurement environment for applied economics.

Measurement model

$$M_i = g(U_i; \theta) + \varepsilon_i$$

Object	Applied interpretation
U_i	raw document, image, audio clip, video, scan, or multimodal record
$g(\cdot)$	dictionary, classifier, embedding, prompt, or multimodal extractor
θ	terms, labels, weights, thresholds, model parameters, prompts
M_i	remote-work score, exposure, risk, poverty proxy, tone, policy measure
ε_i	extraction error relative to the intended economic object

Validation is part of the design because the measure enters later economics.

Extraction families

Family	Best starting point	Main caveat
Dictionaries / rules	explicit terms, clauses, mentions	brittle to wording and negation
Supervised classification	stable labels with human annotation	target definition dominates
Embeddings / similarity	semantic exposure, task matching, document alignment	interpretation can drift
Multimodal features	visual, vocal, or interaction signals	validation burden rises sharply

$$\widehat{f} = \arg\min_{f} \frac{1}{n} \sum_{i} L(y_i, f(x_i)) + \lambda P(f)$$

Dictionary and embedding measures

$$D_i(S) = \frac{1}{N_i} \sum_{t \in U_i} \mathbf{1}\{t \in S\}$$

$$e_i = \frac{1}{N_i} \sum_{t \in U_i} h(t), \qquad S_i(c) = \frac{e_i' c}{\|e_i\| \|c\|}$$

- Dictionary shares are transparent and easy to audit.
- Embeddings broaden beyond exact terms to semantic similarity.
- Both require an economic interpretation before they become variables.
- The same score can mean mention, salience, exposure, or similarity depending on the design.

Text applications

Setting	Constructed variable	Economic use
Job postings	remote-work feasibility	jobs, firms, occupations, places
Patent and task text	AI exposure	technology-labor substitution and complementarity
Earnings calls	epidemic-risk exposure	firm shocks and expectations
Open-ended responses	motives, beliefs, narratives	survey measurement and mechanisms
Contracts / policy text	clauses, rules, stringency	institutions and constraints

Text is powerful because institutions write things down; it is risky because language is strategic and drifting.

Image and satellite applications

- Satellite imagery can extend measurement to places with sparse surveys.
- Street-view imagery can measure built environment, amenities, storefronts, roads, or visible disorder.
- Scanned documents can turn paper archives into structured research inputs.
- Image measures need ground truth: surveys, administrative records, human labels, or known aggregates.

Common failure modes:

- sensor, season, weather, and resolution differences;
- proxy instability across countries or cities;
- visually salient features that are not welfare-relevant;
- domain shift between training and target places.

Audio and video applications

Modality	Possible economic object	Main risk
Audio	tone, stress, fluency, confidence, disclosure	identity and recording artifacts
Speech transcripts	topics, beliefs, narratives	loss of paralinguistic information
Video	interaction quality, sequence, task execution	privacy and high labeling burden
Multimodal records	joint text-image-audio state	unclear mapping and opacity

Use richer modalities only when they measure something the simpler record cannot.

Modality-specific failure modes

Modality	Typical failure	Applied diagnostic
Text	semantic drift, boilerplate, label leakage	time splits, subgroup labels, audit terms
Images	context mismatch and sensor artifacts	place splits, ground truth, image-quality checks
Audio	speaker identity and recording quality	balanced labels, device checks, residual audits
Video	privacy, high-dimensional shortcuts	manual review, ablations, consent documentation
Multimodal	one modality dominates the index	modality ablations and failure examples

The failure mode determines which downstream estimates are most fragile.

Measurement error propagation

$$\widehat{M}_i = M_i^* + \eta_i, \qquad Y_i = \alpha + \beta M_i^* + \nu_i$$

$$\operatorname{plim} \widehat{\beta}_{\widehat{M}} = \beta \cdot \frac{\mathrm{Var}(M_i^*)}{\mathrm{Var}(M_i^*) + \mathrm{Var}(\eta_i)} \quad \text{under classical error}$$

- Classical noise attenuates simple regressions.
- Nonclassical errors can bias estimates in either direction.
- Errors often vary by occupation, firm, region, language, speaker, platform, or sensor.
- If the measure supplies the identifying variation, validation is part of identification.

Research Lab design

Reproduce

- synthetic job postings
- remote-work target
- dictionary score
- embedding similarity
- supervised score

Diagnose

- test metrics
- subgroup errors
- dictionary sensitivity
- leakage audit
- downstream attenuation

Transfer

- policy text records
- image, audio, video features
- modality ablations
- transfer memo

The lab reproduces measurement logic, not official paper estimates.

Takeaways

- Start with the economic object, not the modality or algorithm.
- Choose dictionaries, classifiers, embeddings, or multimodal features based on the mapping problem.
- Validate the constructed measure where it will be used.
- Report failure modes by time, group, geography, platform, and modality.
- Treat measurement error as a threat to downstream economics, not only as a prediction metric.